FLOODGAUGE · FEASIBILITY ASSESSMENT

FloodGauge feasibility assessment

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/floodgauge		
Source	docs/08-feasibility.md @ dbb9a87		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Initial draft for the kit 1.9.1 upgrade

FloodGauge feasibility assessment

Overall verdict: a FloodGauge is feasible as a garage-built pilot at TRL 3 for the physics, parts and assembly, with the radar module data sheet, the drain head seal and road-authority permission as the three things that could stop it. Method: STANDARDS section 23. Ratings describe the recommended pair of options (A or B, FLG-OPT-001); option C is feasible but misses the accuracy target.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Every claim checked is within known science or plausible (15 within, 17 plausible, none fails); the detail model of all three options passes the D3 fidelity check; the only requirements not met on paper are over target on cost and at risk on airtime, immersion and heat	High	Radar accuracy and echo logic rest on an unverified data sheet (STM S3); the 4.994 kg mass margin is 0.1 % and the through-bolt twist factor is 1.8, not 37 (STM S11)	STM, CAL-001, cad/fidelity.json
Materials	Aluminium 6063 arm, galvanised steel covers, stainless fasteners, PVC tube; all stock sections	High	Galvanic contact of stainless with aluminium and zinc in road salt (STM S19)	BOM, STM
Manufacturing and assembly	A two-person crew can make and fit the kit with hand tools; the build plan sequences 10 steps in 120 min on site	Medium	Bent hat covers, V-blocks and cleats need a bench and a brake; the arm is lifted from a platform	BLD-001, CAL-001 L1
Power and controls	9.3 mW worst-case sensor load against a 100 mW allowance; 47 days at -20 C without sun	High	FieldNode interior heat above the cell's 45 C charge limit (PR-014 at risk)	CAL-001 D, FieldNode FND-CAL-001
Supply of parts	Radar module, ultrasonic ranger, stainless fasteners and sections from catalogue suppliers	Medium	Radar module and housing prices are unverified (BOM item 2, USD 55.00 indicative); pressure transducer prices for option B are estimates	BOM, OPT
Regulation and permission	A fixed outdoor 60 GHz radar and an arm over a road need a certificate and a permit	Medium	G1, G2, G9 and G10 are all "to assess"	REG
Cost	USD 205.50 for FloodGauge-specific parts, USD 45.50 over the USD 160 target; option B estimated about USD 191 and option C about USD 171	Medium	The target is a value-engineering control, not a limit	BOM, PR-016, OPT

Ratings: High, Medium, Low.